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Code:

First, create a virtual environment and then install numpy and matplotlib using pip install [library name] in the terminal.

```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image # Image load karne ke liye
# 1. PC se Image Load karein
# 'test_image.jpg' ki jagah apni kisi bhi image ka naam likhein
try:
    img = Image.open('Vango Art.jpg')
    img_array = np.array(img) # Image ko Numpy array mein badal
    diya
except:
    print("Error: Image file nahi mili! File ka naam sahi check
    karein.")
    # Agar file nahi hai to hum practice ke liye random colors
    bana lete hain
    img_array = np.random.randint(0, 255, (300, 300, 3),
    dtype='uint8')

# 2. Image ka structure samjhein
print(f"Shape: {img_array.shape}") # (Height, Width, Channels) -
Channels 3 hote hain (R, G, B)

# 3. Grayscale (Black & White) Conversion Logic
# B&W karne ka formula: (Red + Green + Blue) / 3
# Hum axis=2 use karte hain kyunki RGB channels 3rd dimension mein
hain
grayscale = np.mean(img_array, axis=2)

# 4. Color Filters (Red Filter)
# Hum Green aur Blue channels ko zero kar denge
red_filter = img_array.copy()
```

```

red_filter[:, :, 1] = 0 # Green channel = 0
red_filter[:, :, 2] = 0 # Blue channel = 0

# 5. Visualizing with Matplotlib
plt.figure(figsize=(15, 5)) # Window ka size set kiya

# Original Image
plt.subplot(1, 3, 1) # 1 row, 3 columns, position 1
plt.imshow(img_array)
plt.title("Original RGB")
plt.axis('off')

# Grayscale Image
plt.subplot(1, 3, 2) # position 2
plt.imshow( grayscale, cmap='gray')
plt.title("Black & White (Mean)")
plt.axis('off')

# Red Filtered Image
plt.subplot(1, 3, 3) # position 3
plt.imshow(red_filter)
plt.title("Only Red Channel")
plt.axis('off')

plt.show()

```

Code Concepts Explanation:

```

img_array = np.random.randint(0, 255, (300, 300, 3), dtype='uint8')
print(f"Shape: {img_array.shape}")

```

Yeh line ek “**Fake**” ya “**Dummy**” **colored image** create karne ke liye hai. Let's break down each part step by step:

1. `np.random.randint( 0, 255, ... )`

Yeh function **random integers** (poore numbers) generate karte hai.

- **0**: Sab se choti value (minimum).
- **255**: Sab se bari value (maximum).
- **Logic**: Image ka har pixel 0 se 255 ke beech hota hai, isliye yeh range rakhi.

2. (300, 300, 3) - Yeh sab se important hai:

Yeh array ki **Shape** define kar raha hai:

- **300 (Pehla)**: Image ki **Height** (300 rows).
- **300 (Doosra)**: Image ki **Width** (300 columns).
- **3 (Teesra): Colors** (Channels). Computer mein har color 3 cheezon se banta hain: **Red**, **Green**, and **Blue** (RGB).
- **Logic**: Yeh ek aisi image banayega jo 300x300 pixels ki hogi aur har pixel ke paas 3 random numbers honge (RGB).

3. dtype='uint8'

Yeh data type batata hai:

- **uint8** ka matlab hai “Unsigned integer 8-bit”.
- Iska range 0 se 255 tak hota hai.
- **Logic**: Image hamesha uint8 format mein save hoti hain taake memory kam consume ho aur standard format barkaraar rahe.

4. print(f “Shape: {img_array.shape}”)

Yeh f-string hai jo output main dikhayegi: **Shape: (300, 300, 3)**.

RGB ka “3” asal mein kya hai?

- **3 Channels (RGB)**: Yeh standard “True Color” image hai. Isme har pixel ke liye 3 numbers hote hain: (Red value, Green value, Blue value).
- **1 Channel (Grayscale)**: Agar aap 3 ki jagah 1 kar dein, to image sirf “Black and White” reh jayegi kyunki usme color information khatam ho jayegi.
- **4 Channels (RGBA)**: Kabhi kabhi images mein 4th channel hota hai jisay ‘**A**’ (**Alpha**) kehte hain. Yeh batata hai ke image kitni **Transparent** hai. PNG images mein aksar 4 channels hote hain.

Interesting Concept: The “Byte” Limit (255):

- Computer mein data bits mein hota hai. Ek color channel **8-bit** leta hai.
- $2^8 = 256$. Yani 0 se 255 tak total 256 shades ho sakte hain.
- **Amazing Fact**: Agar aap teeno channels ko multiply karein (256x256x256), to total **16.7 Million** colors bante hain jo ek aam insani aankh dekh sakti hai.

```
grayscale = np.mean(img_array, axis=2)
```

Ek colored pixel mein 3 values hoti hain: [R, G, B]. Black and White (Grayscale) banane ka asan formula hai ke in teeno ka **Average** nikaal liya jaye.

axis = 2 ka kya kaam hai?

Hamara **img_array** 3D hai: (Height, Width, 3).

- **axis = 0** ka matlab hai “Rows” (upar se niche).
- **axis = 1** ka matlab hai “Columns” (left se right).
- **axis = 2** ka matlab hai “Depth” (yani wo 3 RGB values jo har pixel ke andar chupi hain).

Jab aap **np.mean(..., axis = 2)** kehte hain, to Numpy har pixel ke andar jata hai, uske R, G, and B ko uthata hai, unhein jama karke 3 par divide kar deta hai.

- Input: [200, 100, 50] (Ek pixel)
- Output: $(200+100+50) / 3 = 116.6$
- Ab wo pixel sirf ek single number ban gaya. Jab pixel single number ho, to Matplotlib usay “Gray” shade mein dikhata hai.

Output of your image:

Original RGB



Black & White (Mean)



Only Red Channel



Suppose if our image is not found (**FileNotFoundException**), then numpy creates the following dummy image:

